BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Programming Languages and Translators

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How many input rows are needed for three independent Boolean inputs?

2. What does software copyright restrict?
   A) Using any keyboard
   B) Learning to code
   C) Copying without permission
   D) Saving your own work

3. 32 items are packed in boxes of 5. Use DIV and MOD to give the number of full boxes and leftover items.

4. Entry needs a valid ticket AND membership card. "AND is true if just one input is true." Correct this. What happens with only a ticket?

Today's objectives

Explain the core ideas of programming languages and translators using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

High-level language

Human-readable abstraction; generally portable with suitable runtime/translator support.

Low-level language

Close to processor instruction architecture, giving greater hardware control.

Compiler

Translates source code into executable machine or intermediate output before running the resulting program.

Interpreter

Processes/translates and executes code using an interpreter during execution.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

High-level language
Human-readable abstraction; generally portable with suitable runtime/translator support.

Low-level language
Close to processor instruction architecture, giving greater hardware control.

Compiler
Translates source code into executable machine or intermediate output before running the resulting program.

Interpreter
Processes/translates and executes code using an interpreter during execution.

I Do - Worked model

A developer wants one program to run on different processors. Explain how language level and translators matter.

- High-level code is less tied to a specific processor architecture.
- A suitable translator/runtime must still exist for each target.
- Low-level machine code is closely processor-specific.
- Explain why a compiler/interpreter approach is chosen for the environment, not a guaranteed speed ranking.

We Do - Guided practice

1. Compare an interpreter and compiler in detecting errors.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the size calculation for four two-byte characters.

```
characters = 4
bytesPerCharacter = 2
sizeBytes = characters + bytesPerCharacter
print(sizeBytes)
```

You Do - Independent exam questions

Q2. Give one benefit of high-level languages. [2 marks]

Q3. Compare compiler and interpreter at a broad level. [4 marks]

Q4. Explain why machine code depends on processor architecture. [2 marks]

Exit Ticket - True or False?

A high-level program still needs translation/runtime support to execute on a processor.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. 8, because 2 to the power 3 is 8; list every combination exactly once.

2. C - Copying without permission

3. 32 DIV 5 = 6 full boxes; 32 MOD 5 = 2 left over.

4. AND is true only when both inputs are true. One valid ticket or card alone is not enough. OR needs at least one true input; NOT reverses a value.

I Do - Explain the model

High-level code is less tied to a specific processor architecture.

A suitable translator/runtime must still exist for each target.

Low-level machine code is closely processor-specific.

Explain why a compiler/interpreter approach is chosen for the environment, not a guaranteed speed ranking.

We Do - Model answers

1. Do not claim all interpreters execute exactly one source line directly.

Code First - Answer

```
characters = 4
bytesPerCharacter = 2
sizeBytes = characters * bytesPerCharacter
print(sizeBytes)
```

Expected output: 8

You Do - Self-mark

Q2: More readable and often easier to develop/maintain, or portable with suitable runtimes.

Q3: Compiler translates a program to executable/intermediate output; interpreter executes through an interpreting environment.

Q4: Instructions use a machine-specific instruction set understood by that CPU.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.